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## Models

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### Review $y' = A y + b$

1. Write down the complementary solution  $y_c = \sum_{i=1}^n c_i y_i$  satisfying  $y_c' = A y_c$ 
    - 1.1. One solution when  $\lambda$  is real and  $A v = \lambda v$   
 $e^{\lambda t} v$
    - 1.2. Two solutions when  $\lambda = a \pm b i$  and  $A(p + q i) = (a + b i)(p + q i)$   
 $e^{a t}(\cos(b t) p - \sin(b t) q)$  and  $e^{a t}(\sin(b t) p + \cos(b t) q)$
    - 1.3. Two solutions when  $\lambda$  is repeated twice but there is only one eigenvector  $A v = \lambda v$   
 $e^{\lambda t} v$  and  $t e^{\lambda t} v + e^{\lambda t} u$   
where  $u$  satisfies  $(A - \lambda I) u = v$
  2. Compute a particular solution  $y_p$  satisfying  $y_p' = A y_p + b$ 
    - 2.1. Guess the form (polynomial in  $t$ , exponentials  $e^{\omega t}$ ,  $\sin(\omega t)$ , ...) for  $y_p$  if you can and compute the coefficients by substituting into the ODE.
    - 2.2. Use variation of parameters  $y = \Phi(t) u(t)$  where  
 $u(t) = \int^t \Phi(\tau)^{-1} b(\tau) d\tau + c$   
if you can not guess the form.
  3. Write down the general solution  $y = y_c + y_p$ .
  4. Substitute initial conditions (if there are) to determine the vector  $c = [c_1, c_2, \dots, c_n]$ .
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### Example

Solving higher order constant coefficient ODEs e.g.

$$x'' + 2x' + x = t$$

We can convert to a system by introducing  $v = x'$  like this

$$x'' = -2x' - x + t$$

so rearranging gives

$$x' = v$$

$$v' = -x - 2v + t$$

which with  $y = \begin{pmatrix} x \\ v \end{pmatrix}$  is

$$y' = \begin{pmatrix} 0 & 1 \\ -1 & -2 \end{pmatrix} y + \begin{pmatrix} 0 \\ t \end{pmatrix}$$

The matrix has one eigen vector  $v = \begin{pmatrix} -1 \\ 1 \end{pmatrix}$  for a repeated eigenvector of  $\lambda = -1$  so the complementary solution is

$$y_c = c_1 e^{-t} \begin{pmatrix} -1 \\ 1 \end{pmatrix} + c_2 e^{-t} \left( t \begin{pmatrix} -1 \\ 1 \end{pmatrix} + \begin{pmatrix} -1 \\ 0 \end{pmatrix} \right)$$

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In[ ]:= MatrixForm[Eigensystem[A =  $\begin{pmatrix} 0 & 1 \\ -1 & -2 \end{pmatrix}$ ]]
LinearSolve[A - (-1) IdentityMatrix[2], {-1, 1}]
Out[ ]//MatrixForm=
 $\begin{pmatrix} -1 & -1 \\ \{-1, 1\} & \{0, 0\} \end{pmatrix}$ 
Out[ ]=
{-1, 0}
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A particular solution is of the form

$$y_p = u_0 + t u_1$$

where the undetermined coefficients satisfy

$$u_1 = A(u_0 + t u_1) + t \begin{pmatrix} 0 \\ 1 \end{pmatrix}$$

Matching coefficients gives

$$0 = A u_1 + \begin{pmatrix} 0 \\ 1 \end{pmatrix}$$

$$u_1 = A u_0 + \begin{pmatrix} 0 \\ 0 \end{pmatrix}$$

rewriting gives

$$\begin{pmatrix} \begin{pmatrix} 0 & 0 \\ 0 & 0 \end{pmatrix} & -A \\ -A & \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} \end{pmatrix} \begin{pmatrix} u_0 \\ u_1 \end{pmatrix} = \begin{pmatrix} \begin{pmatrix} 0 \\ 1 \end{pmatrix} \\ \begin{pmatrix} 0 \\ 0 \end{pmatrix} \end{pmatrix}$$

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In[ ]:= LinearSolve[ArrayFlatten[ $\begin{pmatrix} 0 & -A \\ -A & \text{IdentityMatrix}[2] \end{pmatrix}$ ]], {0, 1, 0, 0}]
Out[ ]=
{-2, 1, 1, 0}
```

So  $u_0 = \begin{pmatrix} -2 \\ 1 \end{pmatrix}$  and  $u_1 = \begin{pmatrix} 1 \\ 0 \end{pmatrix}$  and the general solution is

$$y = \begin{pmatrix} x \\ v \end{pmatrix} = y_c + y_p = c_1 e^{-t} \begin{pmatrix} -1 \\ 1 \end{pmatrix} + c_2 e^{-t} \left( t \begin{pmatrix} -1 \\ 1 \end{pmatrix} + \begin{pmatrix} -1 \\ 0 \end{pmatrix} \right) + \begin{pmatrix} -2 \\ 1 \end{pmatrix} + t \begin{pmatrix} 1 \\ 0 \end{pmatrix}$$

We might only care about

$$x(t) = -c_1 e^{-t} - c_2(1+t) e^{-t} - 2 + t$$

People would describe this as two decaying transients and a linearly growing particular solution.

## Chapter 4: Teaser

Chapter 4 describes how to solve higher order constant coefficient linear ODEs directly without converting them to a system! The Chapter 4 material matters because it is the way lots of scientists and engineers talk about simple equations such as

$$m x'' + c x' + k x = f_0 \sin(\omega t).$$

This is the standard model for a mass-spring system with mass  $m$ , spring constant  $k$ , and linear damping coefficient  $c$  subject to a sinusoidal (angular frequency  $\omega$ ) force of  $f_0$  Newtons. In the real world, the damping  $c$  is usually small and the angular frequency of the force could be anywhere from slow  $\omega \ll 1$  to fast  $\omega \gg 1$ . We will see for the simple model there is a single angular frequency  $\omega_0$  near which a small force can produce an outsized response. Resonant responses can be disastrous.

## Chapter 4: $y_c$

To find the component solutions of

$$a_2 y'' + a_1 y' + a_0 y = 0$$

simply substitute  $y = e^{\lambda t}$  into the ODE to get

$$a_2 \lambda^2 e^{\lambda t} + a_1 \lambda e^{\lambda t} + a_0 e^{\lambda t} = 0$$

and cancel the exponential to get the characteristic polynomial

$$a_2 \lambda^2 + a_1 \lambda + a_0 = 0$$

Two distinct roots ( $\lambda_1$  and  $\lambda_2$ ) of this quadratic will give the complementary solution

$$y_c = c_1 y_1 + c_2 y_2 = c_1 e^{\lambda_1 t} + c_2 e^{\lambda_2 t}$$

Euler's identity

$$e^{(a+ib)t} = e^{at}(\cos(bt) \pm i \sin(bt))$$

splits complex root pairs  $\lambda = a \pm ib$  into two independent real solutions

$$y_1 = e^{at} \cos(bt)$$

$$y_2 = e^{at} \sin(bt)$$

For a higher order ODE

$$\sum_{i=3}^n a_i y^{(i)} + a_2 y'' + a_1 y' + a_0 y = 0$$

$n$  roots  $\lambda_1, \lambda_2, \dots, \lambda_n$  of the characteristic polynomial would give  $n$  solutions

$$y_i = e^{\lambda_i t}$$

to build the complementary solution

$$y_c = \sum_{i=1}^n c_i y_i.$$

Multiplying by  $t$  generates needed solutions for repeated roots. If  $\lambda$  is repeated twice

$$y_1 = e^{\lambda t} \text{ and } y_2 = t e^{\lambda t}$$

are solutions. If  $\lambda$  is repeated three times then

$$y_1 = e^{\lambda t}, y_2 = t e^{\lambda t} \text{ and } y_3 = t^2 e^{\lambda t}$$

are solutions.

## Chapter 4: $y_p$ undetermined coefficients

For

$$a_2 y'' + a_1 y' + a_0 y = f$$

we can guess solution forms for simple right hand sides  $f$ .

- If  $f$  is a polynomial guess a polynomial of the same order.
- If  $f$  is an exponential guess the same type of exponential.
- If  $f$  is a sine and/or cosine guess a sum of sines and cosines

Write down the form with appropriately named coefficients. Plug the form into the ODE to get equations to determine the coefficients.

The same process works for higher order ODEs. A similar process works for any RHS that you get the same sort of terms when you differentiate.

## Chapter 4: General Solution and IC

The general solution of

$$\sum_{i=0}^n a_i y^{(i)} = f$$

is the sum of a particular solution and the complementary solution

$$y = y_p + y_c = y_p + \sum_{i=0}^n c_i y_i(t)$$

Plugging an appropriate set of initial conditions e.g. values for

$$y(0), y'(0), y''(0), \dots, y^{(n-1)}(0)$$

in gives  $n$  linear equations in the  $n$  unknowns  $c_1, c_2, \dots, c_n$ . Solving this linear system gives you the solution satisfying the initial conditions.